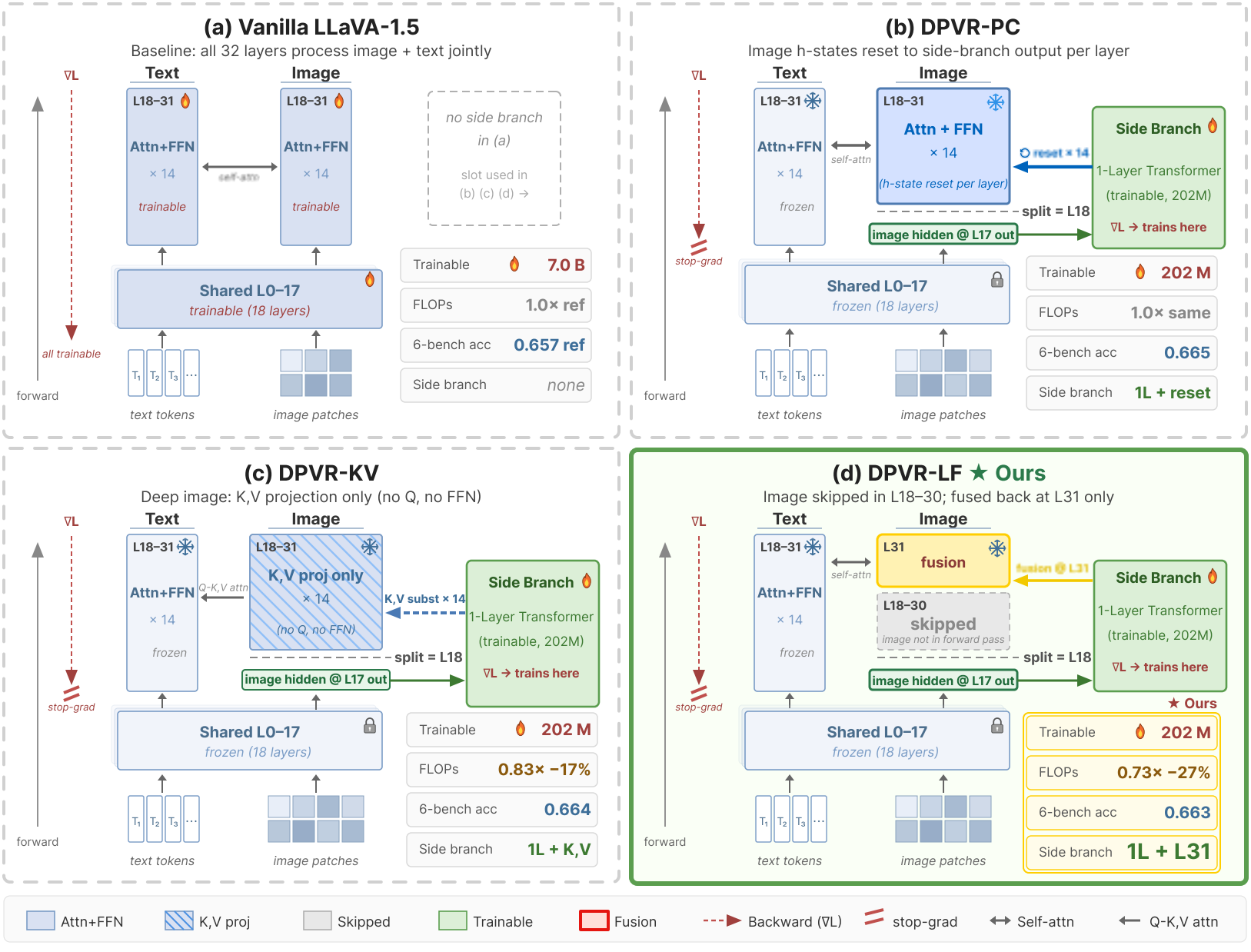

(a) Vanilla LLaVA-1.5
Baseline: all 32 layers process image + text jointly
∇L
Text
Image
L18–31
Attn+FFN
× 14
trainable
self-attn
no side branch
in (a)
slot used in
(b) (c) (d) →
Shared L0–17
trainable (18 layers)
all trainable
forward
text tokens
image patches
Trainable 7.0 B
FLOPs 1.0× ref
6-bench acc 0.657 ref
Side branch none
(b) DPVR-PC
Image h-states reset to side-branch output per layer
frozen
Attn + FFN
(h-state reset per layer)
reset × 14
Side Branch
1-Layer Transformer
(trainable, 202M)
∇L → trains here
split = L18
image hidden @ L17 out
stop-grad
Shared L0–17
frozen (18 layers)
Trainable 202 M
FLOPs 1.0× same
6-bench acc 0.665
Side branch 1L + reset
(c) DPVR-KV
Deep image: K,V projection only (no Q, no FFN)
K,V proj only
(no Q, no FFN)
Q-K,V attn
K,V subst × 14
Trainable 202 M
FLOPs 0.83× −17%
6-bench acc 0.664
Side branch 1L + K,V
(d) DPVR-LF ★ Ours
Image skipped in L18–30; fused back at L31 only
L31
fusion
fusion @ L31
L18–30
skipped
image not in forward pass
★ Ours
Trainable 202 M
FLOPs 0.73× −27%
6-bench acc 0.663
Side branch 1L + L31
Attn+FFN
K,V proj
Skipped
Trainable
Fusion
Backward (∇L)
stop-grad
Self-attn
Q-K,V attn